

Adaptive Trial Design for the RAS-Inhibitor Era: Combinations, Resistance, and Biomarker-Guided Therapy

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Today's argument

New RAS therapies in PDAC need new kinds of trials.

We know how to **optimize combination doses**.

We know how to **adapt trials to emerging resistance**.

The challenge is making these designs practical enough to launch.

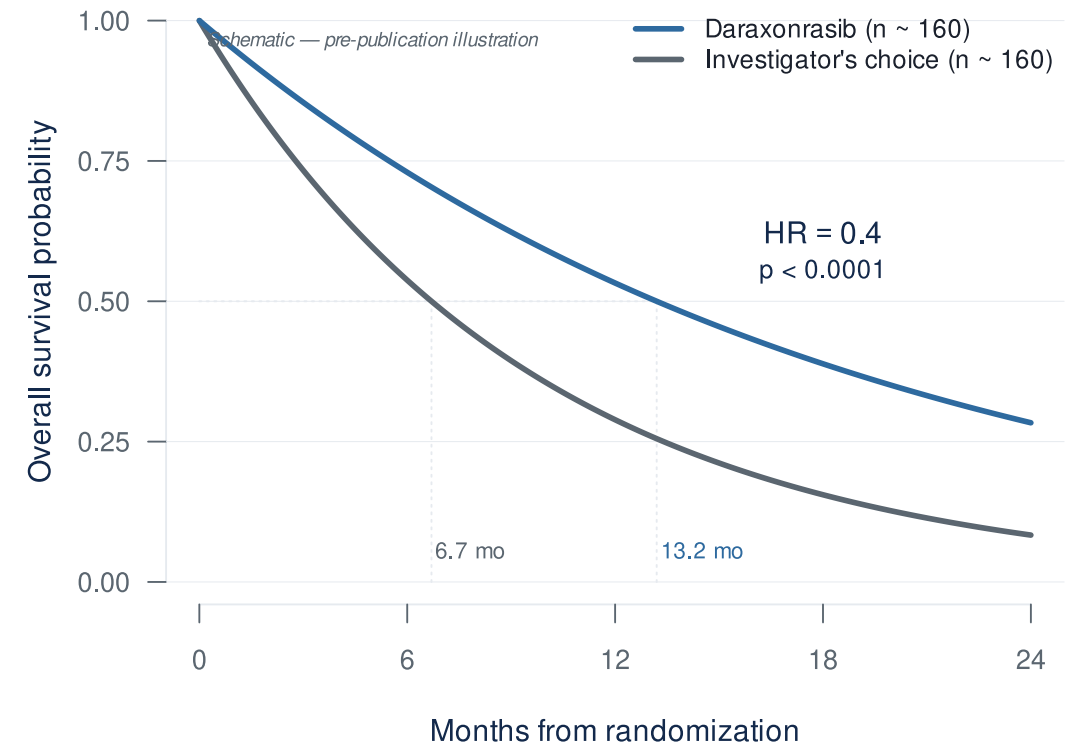
The clinical case

Why combinations, and why now?

The RAS revolution has arrived for PDAC

A major survival signal from direct RAS inhibition in PDAC.

- **RASolute 302 (2L mPDAC, *NEJM* 2026)¹:**
 - **Median OS: 13.2 vs 6.7 months**
 - **HR 0.40 — 60% reduction in risk of death ($p < 0.0001$)**
- Daraxonrasib (RMC-6236) — oral, multi-selective RAS(ON) inhibitor
- ASCO 2026 Plenary, May 31 (Wolpin)



...but monotherapy has visible ceilings

RMC-6236-001 (PDAC cohort, NEJM 2026)²

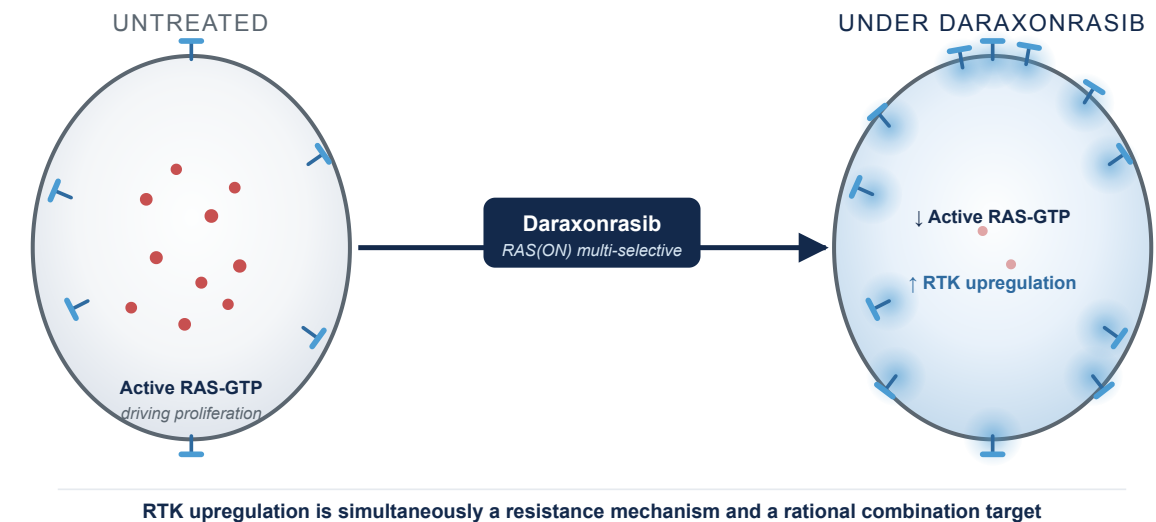


You're combining into a toxicity budget that's already nearly spent. 48% of monotherapy patients already require dose modification at 300 mg. A combination doesn't just add toxicity — it adds **overlapping, non-additive** toxicity that single-agent data can't predict.

Finding the right combination dose becomes the next challenge.

Biomarker-driven combinations are the next move

- Daraxonrasib drives **RTK upregulation** on PDAC cell surface — the mechanistic case for biomarker-driven, resistance-preemptive combinations³
- **1L combination cohort** (daraxonrasib 200 mg + GnP, n=40)⁴:
 - ORR: **58%** · DCR: **90%** · 6-mo PFS: **84%** · 6-mo OS: **90%**



The same adaptation that creates resistance may create a therapeutic vulnerability.

The dose-finding problem

Why MTD only may not be best for RAS combinations, and what replaces it.

Why combinations break the rules you grew up on

- **Rivière's review of 162 combination Phase I trials: 88% used 3+3** even when both drugs were escalated^{5,6}.
- **Why 3+3 doesn't fit combinations.** Built for single-agent cytotoxics⁷, it assumes monotonic dose-toxicity, Cycle-1-observable DLTs, and a single dose dimension — combinations strain all three.
- **The two-drug dose map is never built.** In practice: pick A's dose, fix it, add B. Two monotherapy escalations run in series; the joint efficacy/toxicity behavior across (A, B) pairs (the “joint surface”) is never characterized.

Two-drug dose map: how efficacy and toxicity change across every (A, B) dose pair — not just along one drug's axis.

The Amgen case: sotorasib + pembrolizumab

A real-world example of why combination toxicity can surprise us.

Same sponsor, two different designs:

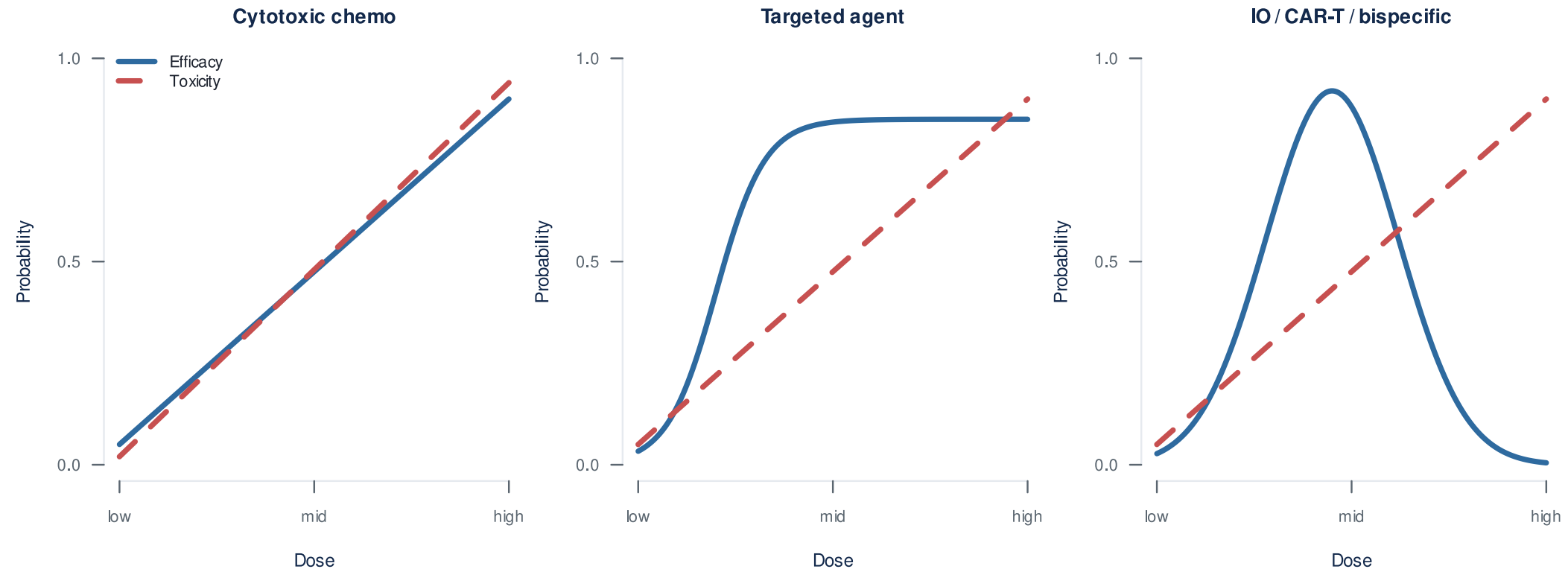
- **Monotherapy (CodeBreakK 100):** Bayesian model-based, single-agent surface
- **Combination:** IO dose fixed, only sotorasib varied — joint surface never modeled

The defining toxicity:

- Severe **immune-mediated** hepatotoxicity, not predictable from the single agent
- **88%** of Grade 3-4 events outside the Cycle 1 DLT window⁸

What MTD-only dose-finding was never built to handle, all at once: joint surface unsearched, non-additive toxicity, late-onset toxicity.

Why MTD-only dose-finding doesn't fit targeted agents and immunotherapies



The cost of MTD-only dose-finding

- Sotorasib (KRAS G12C, first-in-class, approved 2021)¹⁰:
 - 960 mg selected as the **highest dose tested**, no DLTs — a more-is-better default, not an MTD
- **MTD $\neq$ optimal dose** for targeted agents and immunotherapies
- **More than 40% of patients** in small-molecule registrational trials require dose reductions or interruptions¹¹

240 mg $\approx$ 960 mg on PFS

*FDA-required post-marketing comparison: similar PFS at **one-quarter** the exposure¹².*

Same drug class as daraxonrasib.

Project Optimus: the regulatory ground has shifted

How OBD selection actually works

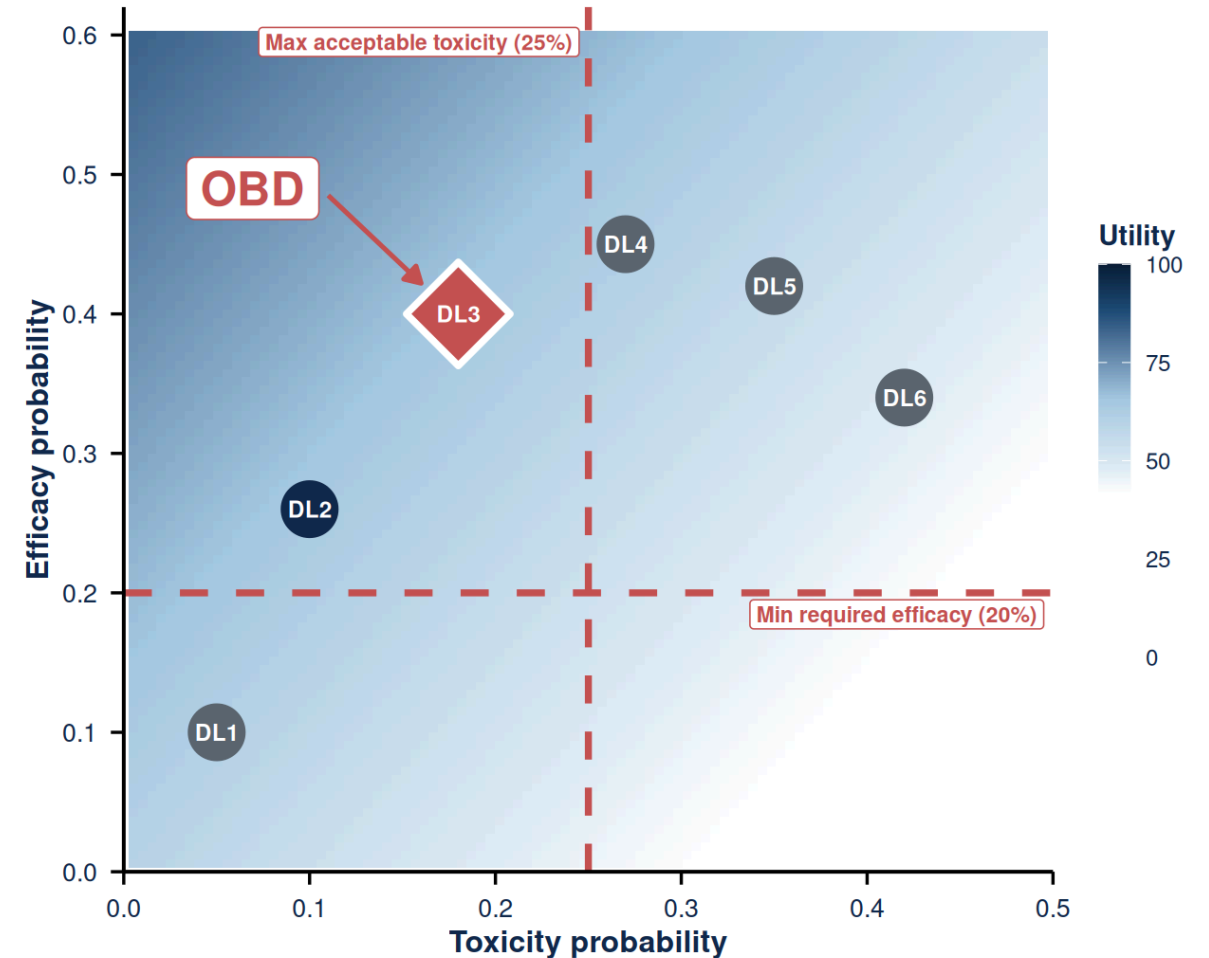
Instead of escalating until toxicity, choose the dose with the best benefit-risk profile.

“How much efficacy would you trade for additional toxicity?”

The team writes that utility table down up front. The design then searches inside the safe envelope for the dose that **maximizes benefit-risk**, not toxicity.

Elicited utility table (Zhou 2019 framework)¹⁴:

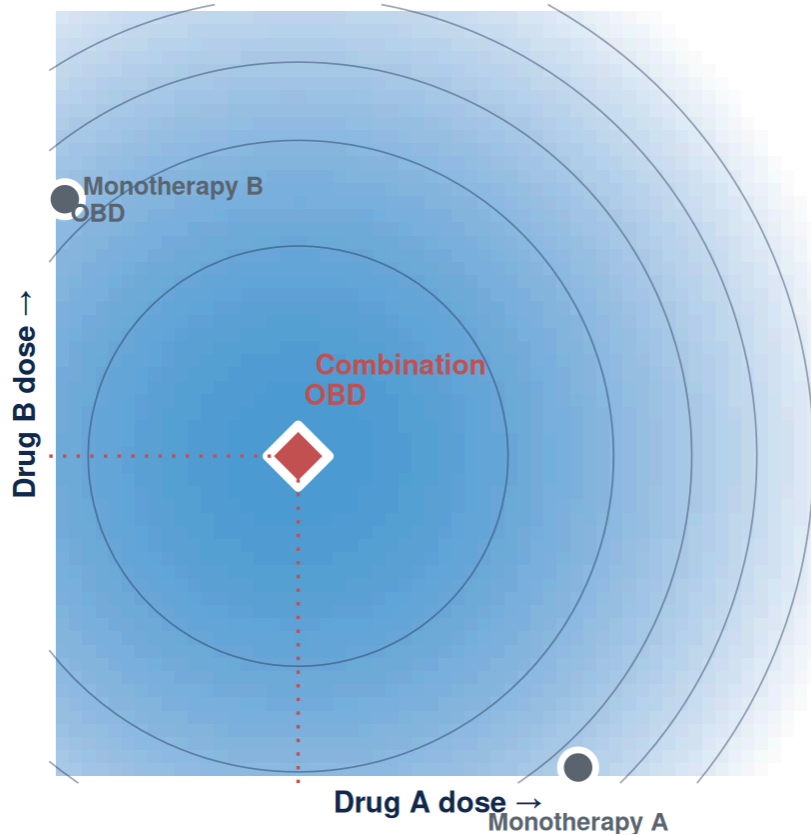
	TOX	NO TOX
Response	40	100
No response	0	60



OBD = admissible dose (inside red lines) with the highest utility.

Combination dose selection in RAS is harder

Combinations make MTD-only dose-finding worse on every axis.



- The combination OBD can sit **below both single-agent OBDs**
- Single-agent data **cannot reconstruct** the joint surface
- For RAS, the **toxicity margin is already spent** at the monotherapy dose

Fix what you know. Search what you don't.

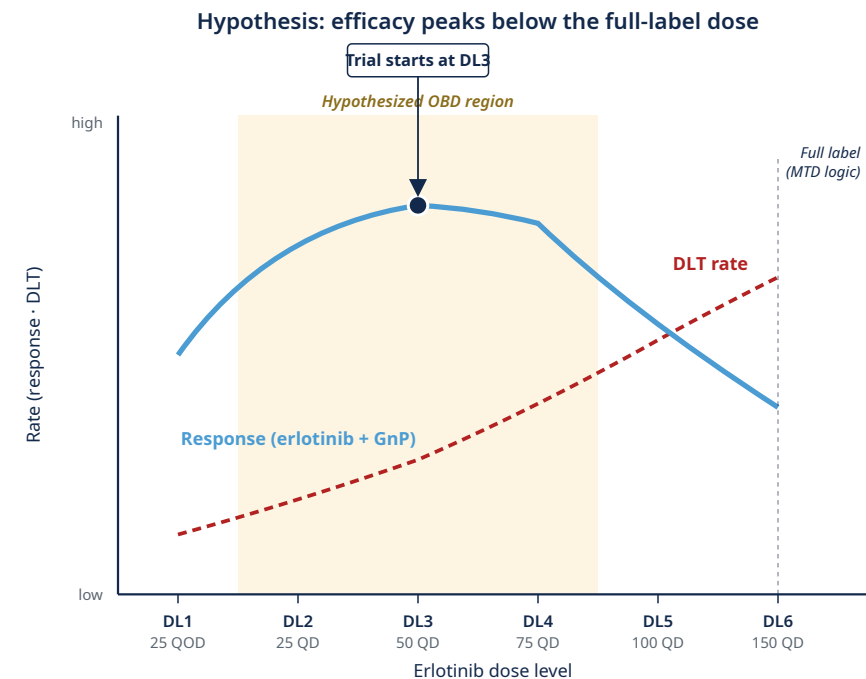
RMC-GI-102 (daraxonrasib + GnP) fixed daraxonrasib at 200 mg without searching the joint surface⁴.

Conceptual schematic of the benefit-risk surface. The combination OBD sits below both single-agent OBDs; single-agent data alone cannot reconstruct the contour.

Finding the right erlotinib dose in basal-like PDAC

- **Clinical question.** Could *lower*-dose erlotinib work *better* than full-dose erlotinib in basal-like PDAC, when combined with GnP?
- **Design.** First find the safe dose range, then choose the dose with the best response/toxicity trade-off^{14,17}.
- **Why it matters.** This is an OBD-below-MTD question; 3+3 cannot answer it.

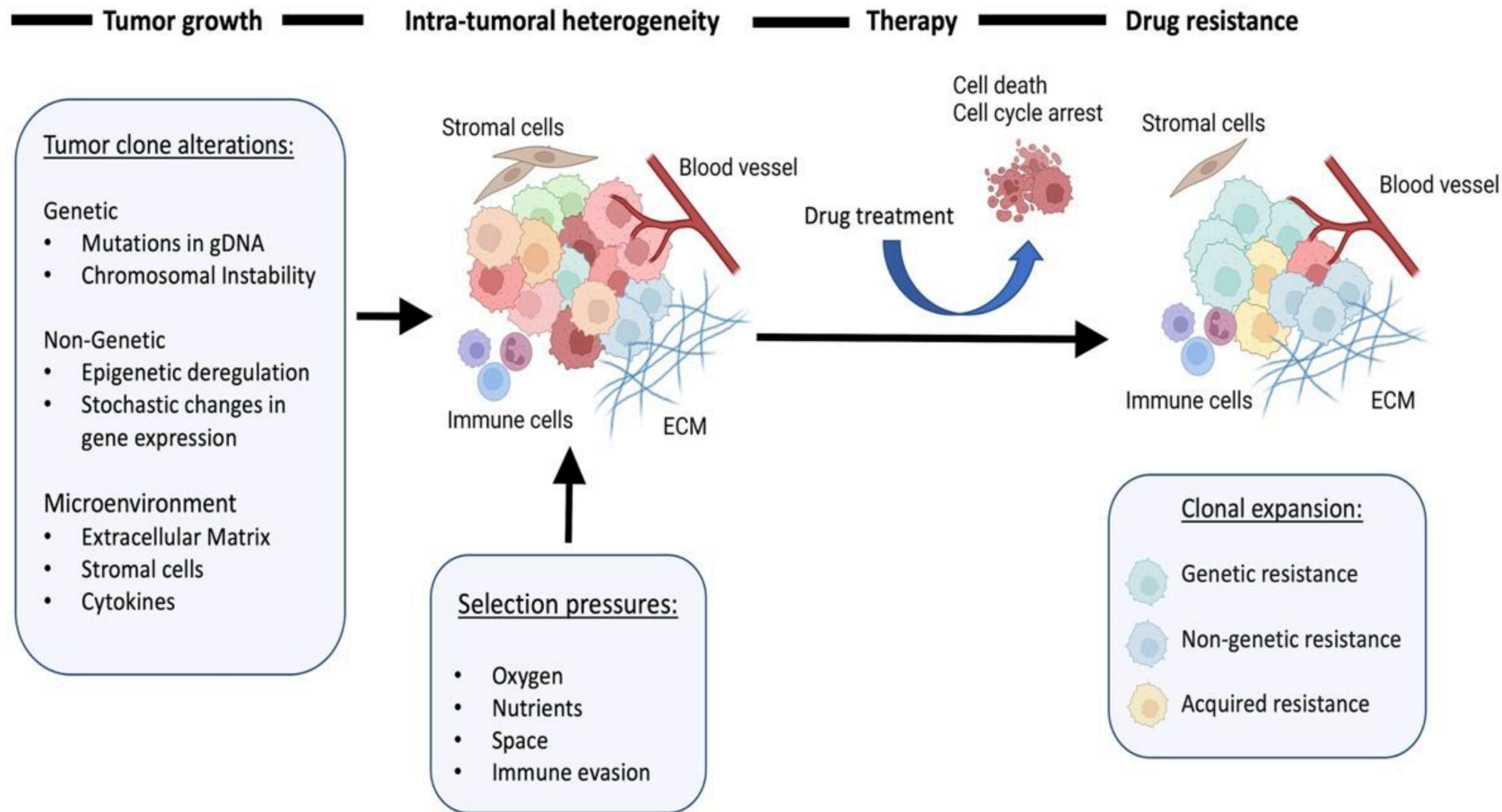
Trial: LCCC2220-DCT (Somasundaram, PI), basal-like PDAC by PurlIST¹⁸ . combination erlotinib + biweekly GnP · OBD-below-MTD hypothesized from prior data¹⁹ · **up to 52 patients; ~28 expected at the OBD.**



Resistance is the next problem

Phase 2 onward: each patient is a moving target

Resistance is a dynamic process

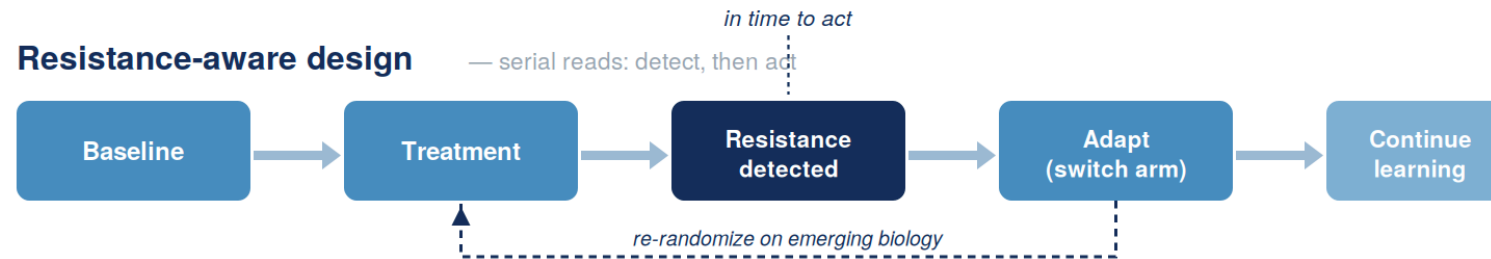


Resistance mechanisms emerge under RAS-inhibitor pressure

Daraxonrasib monotherapy in mPDAC (n=44, paired ctDNA): 59% show RAS-pathway resistance at progression — KRAS amplification in 36%²¹

Adapted from Karagiannis & Rampias, Cancers 2022²⁰.

Why classical trial designs miss resistance



A static read measures a moving target.

Longitudinal detection + biomarker-guided arm-swaps keep the trial in step with the tumor.

RESISTANCE TYPE

WHAT YOU NEED

Genomic (KRAS amplification, NRAS/BRAF/MAP2K1)

ctDNA

Cell-state / plasticity (basal ↔ classical, partial EMT)

biopsy + transcriptomics + imaging

ADAPT / EVOLVE: what a resistance-aware trial looks like

ARPA-H's resistance-tracking template.

ARPA-H's flagship adaptive resistance-monitoring platform.

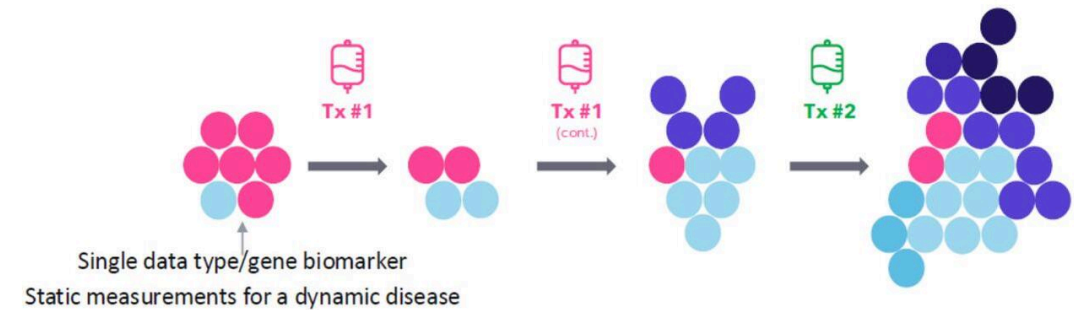
- Longitudinal resistance detection
- Biomarker-guided adaptation
- Adaptive arm-swaps on emerging biology

ARPA-H ADAPT → EVOLVE (UNC-led; Carey MPI, Rashid MPI). Enrolling.

ADAPT Goal: Identify and target changes in metastatic cancers

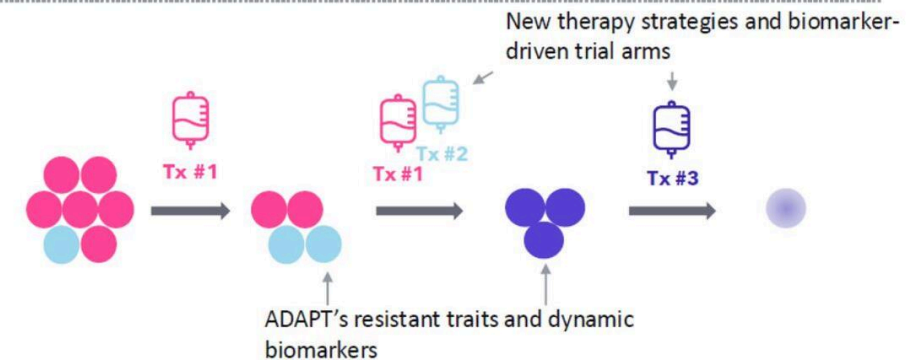
Current State

- Cancer changes during treatment. Therapies **not** often informed by tumor change to cancer resistance traits.
- There are only **limited** tumor measurement biomarker analyses, which are most often performed **prior to** treatment.



ADAPT's upcoming "future state"

- Therapies **will be** informed by and tailored to observed cancer resistance traits using biomarkers.
- **In-depth** tumor measurements will be taken **before, during and after** treatment.



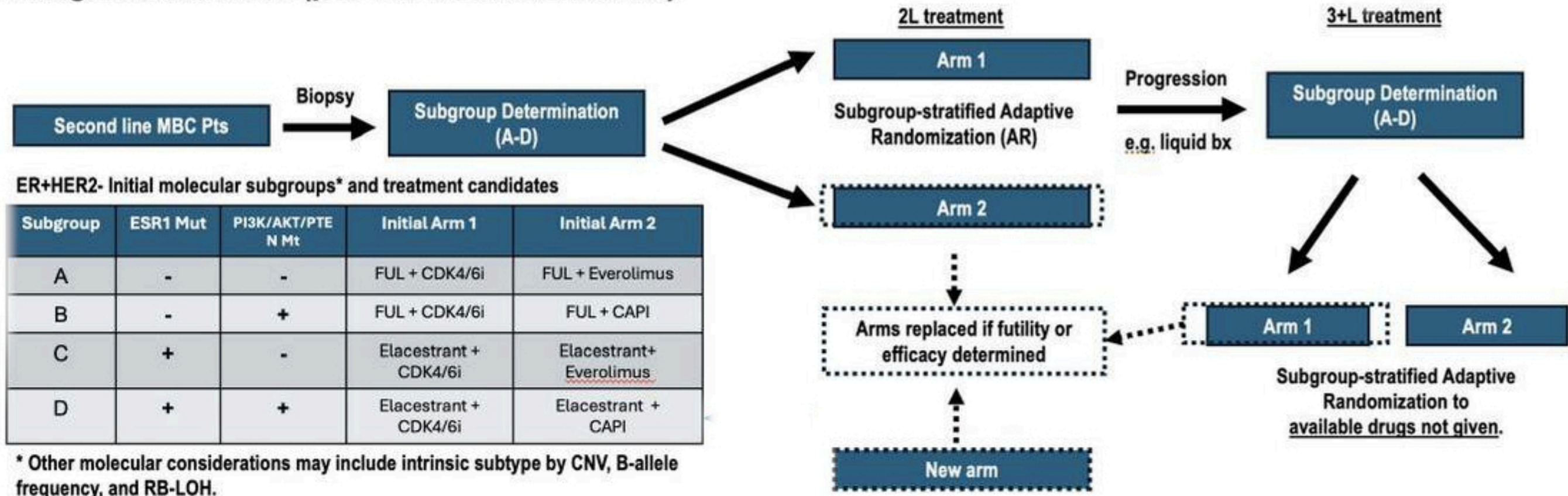
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ARPA-H ADAPT — current state (top) vs dynamic-biomarker state (bottom).

EVOLVE-BDT — and what this would look like in PDAC

EVOLVE is the operational template; PDAC is the next application.

First generation trials (pre-TA1 novel biomarkers)



TBCRC EVOLVE-BDT flow — biomarker-stratified subtrials, adaptive randomization, in-trial arm replacement on futility or efficacy.

EVOLVE-BDT (real, enrolling)

ER+/HER2– and TNBC, biomarker-stratified subtrials with Bayesian arm-swap on futility/efficacy.

PDAC analog (hypothetical)

Basal-like vs classical by PurlST¹⁸, same Bayesian arm-swap logic on daraxonrasib-combination arms.

The same adaptive framework could be applied to PurlST-defined PDAC subtypes and evolving resistance biology.

What it would take to actually run these trials

Four things any such trial has to do at once.

**Stop ineffective
therapies
earlier**

Bayesian futility /
efficacy boundaries
at each interim

**Learn who benefits
while the trial
is running**

Pre-specified signatures
+ agnostic
discovery window

**Track resistance
as it
emerges**

Serial ctDNA,
imaging, biopsies
on every patient

**Test several
biomarker groups
under one protocol**

Biomarker-guided
cohorts within the
same protocol

Everything I've just described already exists statistically. So why don't we see these trials everywhere?

Putting these into practice

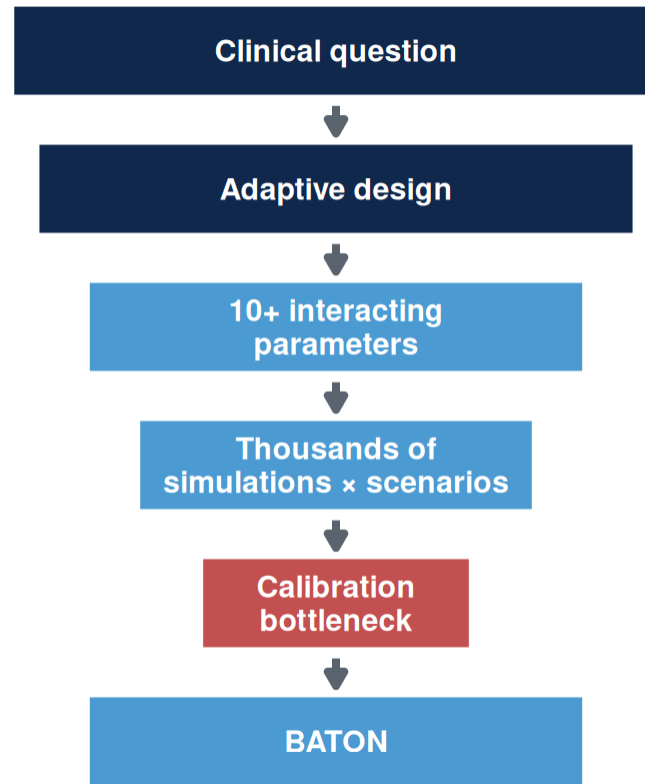
Why don't we see more of these trials?

Funding, accrual, and biopsy logistics are real.

But the barrier that quietly kills good designs — and the one we can actually remove — is calibration.

Before an adaptive trial can open, **thousands of simulations** are needed to prove it is safe, efficient, and scientifically reliable.

That process — not the design itself — is often the bottleneck.



Bayesian designs adopted 75%

Dose-optimization plans 30%

Methods adopted. Dose-optimization plans — where calibration cost bites — not²².

Our group: several months per design. For combined dose + resistance designs, hand-tuning is infeasible.

Calibration has to hit four goals at once

Designing an adaptive trial requires achieving all four simultaneously.

1. FDA compliance

Power ≥ 0.80
Type I ≤ 0.10

2. Minimize patients

Smaller $E[N]$
Efficient use of
scarce patients

3. Stop bad early

Futility stopping
Protect patients
from inactive arms

4. Accelerate good

Efficacy stopping
Faster access
when drug works

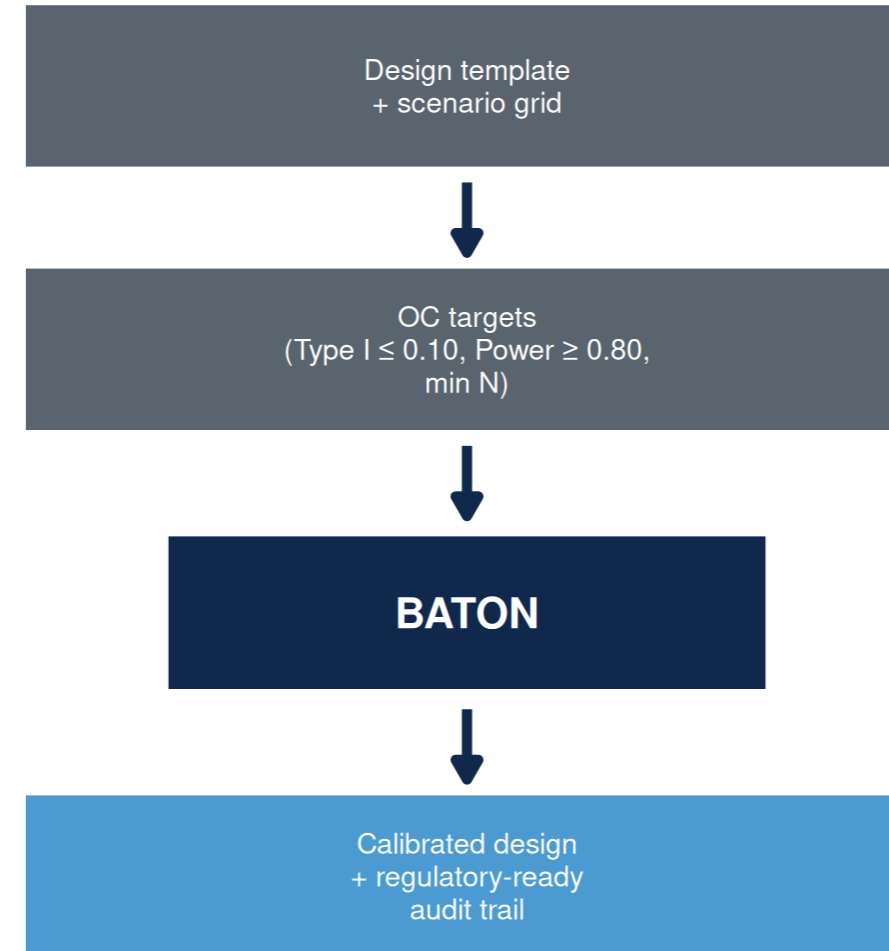
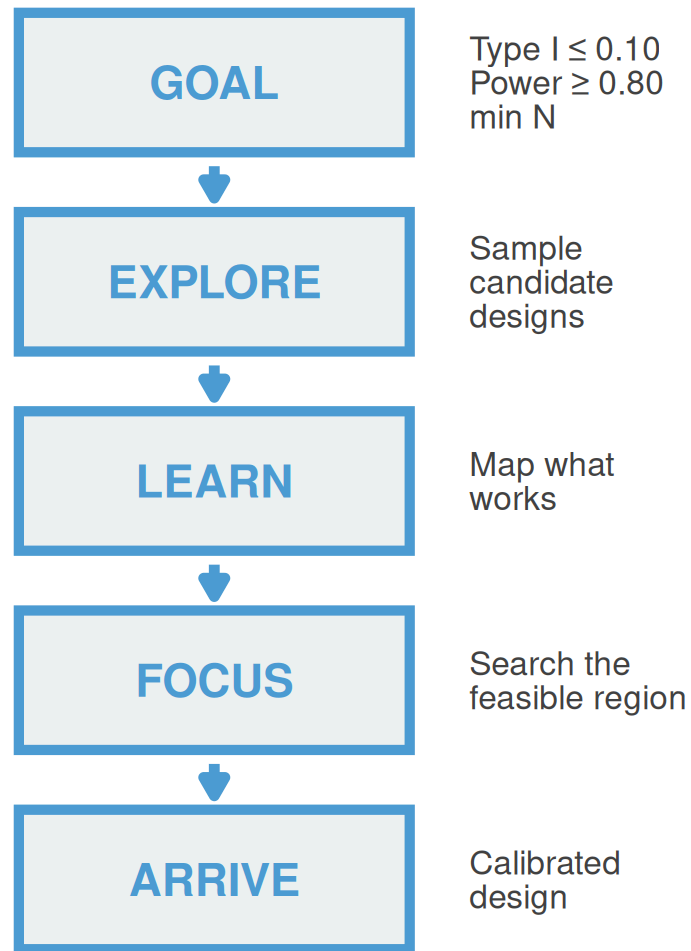
Of thousands of plausible designs, only a small fraction satisfies all four — and you cannot find it by hand.

***“We knew what trial to run.
We needed a way to make it
launchable.”***

BATON: a way to make adaptive trials launchable

BATON automatically searches thousands of candidate designs and identifies those that meet FDA, efficiency, and scientific goals.

BATON = Bayesian Adaptive Trial Optimization²³



You bring the clinical inputs; BATON returns the calibration.

You bring the clinical design; BATON evaluates hundreds of candidates by simulation and returns the ones that meet your targets. R package: github.com/naimurashid/BATON²³

A BATON-calibrated design hits all four goals — for PANGEA Phase II

1. FDA compliance ✓

Type I 7.0% (target $\leq 10\%$)
Power 82.2% (target $\geq 80\%$)

2. Minimize patients ✓

~23 patients spared
(avg N 53 vs Max N 76)

3. Stop bad early ✓

78% early futility stop
when drug is inactive

4. Accelerate good ✓

81% early efficacy stop
when drug works

Far fewer evaluations than grid search — finds feasible designs that grid search often misses entirely. For a 4-parameter design like this one: manual calibration takes days, BATON takes minutes, with a regulatory-ready audit trail.

What this means for a PDAC patient

Mr. Hernandez, 62, mPDAC, basal-like by PurlST, progressed on 1L GnP.

Traditional fixed Phase II

- All **76 patients** enrolled before any look
- Drug effect discovered only at study end
- **~36+ months** to read out

PANGEA Phase II (BATON-calibrated)

- **78% chance** of early futility stop **when the drug is inactive**
- Average patients enrolled: **~53** if inactive / **~58** if active
- **27.3 months** expected duration

Sooner answer, fewer patients on a losing arm — without losing power.

PANGEA Phase II: calibrated by BATON

BATON identified a design that preserves power, controls false positives, and allows early stopping when erlotinib isn't helping.

- After Phase I OBD → **Phase II Bayesian adaptive RCT**: GnP + erlotinib (at OBD) vs GnP alone, primary PFS
- **Design assumes** reference 5.5 mo → 9.2 mo (HR 0.58)²⁴
- **BATON-calibrated** four parameters under Type I ≤ 0.10 , Power ≥ 0.80 ²³

Achieved operating characteristics:

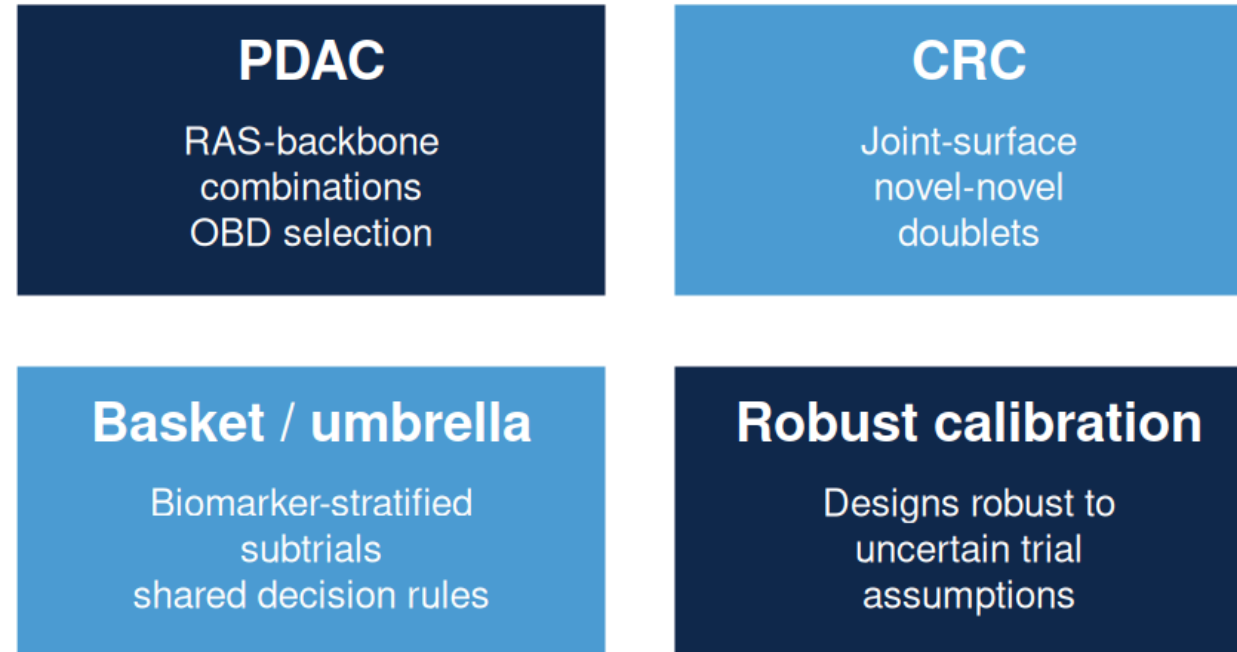
METRIC	VALUE
Power	82.2%
Type I error	7.0%
Max N	76 (82 accrual w/ buffer)
Expected N under H_0	53.0
Expected N under H_1	58.1
Pr(early futility stop H_0)	77.9%
Pr(early efficacy stop H_1)	81.4%
Expected duration	27.3 mo

H_0 = drug inactive · H_1 = drug active (HR 0.58). Read as: 78% early-stop when inactive; 81% early-stop when it works.

Why max N = 76, not lower: caps set too low silently suppress early-efficacy stopping. BATON surfaced the trade-off curve; we chose 76 to preserve it.

What becomes possible

BATON calibrates **aggressive**, **conservative**, or **balanced** designs — and ports across GI applications:



BATON is the infrastructure that makes adaptive trial design routine across the GI portfolio.

Adaptive, resistance-aware trials as the routine standard across GI — not the exception.

Take-home

1. **The RAS era needs new kinds of trials** — biomarker-guided, dose-optimized, resistance-aware.
2. **The methods exist.** Bayesian adaptive designs can handle dose optimization, biomarker guidance, and resistance-aware adaptation.
3. **BATON removes one of the major barriers preventing these designs from reaching patients.**

Acknowledgments

Clinical collaborators

- Lisa Carey (UNC, EVOLVE PI)
- Channing Der (UNC, NEJM editorial²⁵)
- Jen Jen Yeh (UNC Surgery, STOP SPORE PI · NEJM editorial · PDAC collaborator)
- Ashwin Somasundaram (PANGAEA PI)

EVOLVE MPI team

- Charles Perou (UNC) · Ian Krop, Eric Winer (Yale) · Antonio Wolff (Hopkins)

Methods collaborators

- Stephen Bates (MIT) · Ying Yuan (MD Anderson) · Qian Cao (FDA CDRH)

EVOLVE Statistical Working Group

- Susan Hilsenbeck (Baylor) · Nabihah Tayob (DFCI) · Ruizhe Chen (Hopkins)

Rashid Lab

- **Amber Young**, Tyler Humpherys, Dinelka Nanayakkara, Jialiu Xie, Andrew Walther

Funding

- **NCI P50-CA257911** — UNC STOP SPORE
- ARPA-H 140D042590009
- DOD HT9425241103100121
- NCI U01-CA274298

References

1. Eileen M. O'Reilly, Zev A. Wainberg, Andrew E. Hendifar, Mitesh J. Borad, Filippo Pietrantonio, Shubham Pant, Pascal Hammel, Chiara Cremolini, Gulam A. Manji, Paul E. Oberstein, Ignacio Garrido-Laguna, Christoph Springfeld, Nilofer S. Azad, Makoto Ueno, Stephen Y. Chui, Ying Zhang, Hina Patel, Yeonju Lee, Zeena Salman, Brian M. Wolpin, RASolute 302 Trial Investigators. Daraxonrasib or chemotherapy in previously treated metastatic pancreatic cancer. *New England Journal of Medicine*. 2026. doi:[10.1056/NEJMoa2605555](https://doi.org/10.1056/NEJMoa2605555)
2. Brian M. Wolpin, others. Daraxonrasib in previously treated advanced RAS-mutated pancreatic cancer. *New England Journal of Medicine*. 2026;394(18):1790–802. doi:[10.1056/NEJMoa2505783](https://doi.org/10.1056/NEJMoa2505783)
3. Yongxian Zhuang, others. The RAS(ON) multi-selective inhibitor daraxonrasib (RMC-6236) induces receptor tyrosine kinase cell-surface expression on pancreatic cancer cells. *Cancer Research*. 2026;86(5_Suppl_1):PR009. doi:[10.1158/1538-7445.RASONCOTHER26-PR009](https://doi.org/10.1158/1538-7445.RASONCOTHER26-PR009)
4. Brian M. Wolpin, Benjamin L. Musher, Gulam A. Manji, others. Daraxonrasib plus chemotherapy as first-line treatment for patients with metastatic pancreatic adenocarcinoma. *AACR Annual Meeting 2026*, Abstr LB407 (RMC-GI-102, NCT06445062); 2026.
5. Marie-Karelle Rivière, Christophe Le Tourneau, Xavier Paoletti, Frédéric Dubois, Sarah Zohar. Designs of drug-combination phase I trials in oncology: A systematic review of the literature. *Annals of Oncology*. 2015;26(4):669–74. doi:[10.1093/annonc/mdu516](https://doi.org/10.1093/annonc/mdu516)

6. Jennifer A. Harrington, Graham M. Wheeler, Michael J. Sweeting, Adrian P. Mander, Duncan I. Jodrell. Adaptive designs for dual-agent phase I dose-escalation studies. *Nature Reviews Clinical Oncology*. 2013;10(5):277–88. doi:[10.1038/nrclinonc.2013.35](https://doi.org/10.1038/nrclinonc.2013.35)
7. Christophe Le Tourneau, J. Jack Lee, Lillian L. Siu. Dose escalation methods in phase I cancer clinical trials. *Journal of the National Cancer Institute*. 2009;101(10):708–20. doi:[10.1093/jnci/djp079](https://doi.org/10.1093/jnci/djp079)
8. Bob T. Li, Gerald S. Falchook, Gregory A. Durm, others. OA03.06 CodeBreak 100/101: First report of safety/efficacy of sotorasib in combination with pembrolizumab or atezolizumab in advanced KRAS p.G12C NSCLC. *Journal of Thoracic Oncology*. 2022;17(9):S10–1.
9. Amita Patnaik, others. Phase I study of pembrolizumab (MK-3475; anti-PD-1 monoclonal antibody) in patients with advanced solid tumors. *Clinical Cancer Research*. 2015;21(19):4286–93. doi:[10.1158/1078-0432.CCR-14-2607](https://doi.org/10.1158/1078-0432.CCR-14-2607)
10. David S. Hong, others. KRAS^{G12C} inhibition with sotorasib in advanced solid tumors. *New England Journal of Medicine*. 2020;383(13):1207–17. doi:[10.1056/NEJMoa1917239](https://doi.org/10.1056/NEJMoa1917239)
11. Hao Zhu, Alex N. Phipps, Ying Yuan, Brad A. Davidson, Stacy S. Shord, Jiang Liu, Patricia M. LoRusso. FDA–AACR strategies for optimizing dosages for oncology drug products: Selecting dosages for first-in-human trials. *Clinical Cancer Research*. 2025;31(23):4872–81. doi:[10.1158/1078-0432.CCR-25-0095](https://doi.org/10.1158/1078-0432.CCR-25-0095)
12. Maximilian J. Hochmair, Karim Vermaelen, Giannis Mountzios, others. Sotorasib (960 mg or 240 mg) once daily in patients with previously treated KRAS G12C-mutated advanced NSCLC. *European Journal of Cancer*. 2024;208:114204. doi:[10.1016/j.ejca.2024.114204](https://doi.org/10.1016/j.ejca.2024.114204)

13. U.S. Food and Drug Administration. Optimizing the dosage of human prescription drugs and biological products for the treatment of oncologic diseases. Guidance for Industry (final); 2024.
14. Yanhong Zhou, J. Jack Lee, Ying Yuan. A utility-based bayesian optimal interval (U-BOIN) phase I/II design to identify the optimal biological dose for targeted and immune therapies. *Statistics in Medicine*. 2019;38(28):5299–316.
15. Ruitao Lin, Yanhong Zhou, Fangrong Yan, Daniel Li, Ying Yuan. BOIN12: Bayesian optimal interval phase I/II trial design for utility-based dose finding in immunotherapy and targeted therapies. *JCO Precision Oncology*. 2020;4:1393–402.
16. Peter F. Thall, Elizabeth Garrett-Mayer, Nolan A. Wages, Susan Halabi, Ying Kuen Cheung. Current issues in dose-finding designs: A response to the US FDA oncology center of excellence project optimus. *Clinical Trials*. 2024;21(3):267–72.
17. Suyu Liu, Ying Yuan. Bayesian optimal interval designs for phase I clinical trials. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*. 2015;64(3):507–23.
18. Naim U. Rashid, Xianlu L. Peng, Chong Jin, Richard A. Moffitt, Keith E. Volmar, Brian A. Belt, Roheena Z. Panni, Timothy M. Nywening, Stephen G. Herrera, Kathleen J. Moore, Sharon G. Hennessey, Ashley B. Morrison, Ryan Kawalerski, Apoorve Nayyar, Albert E. Chang, Bradley Schmidt, Hong Jin Kim, David C. Linehan, Jen Jen Yeh. Purity independent subtyping of tumors (PurIST): A clinically robust, single-sample classifier for tumor subtyping in pancreatic cancer. *Clinical Cancer Research*. 2020;26(1):82–92. doi:[10.1158/1078-0432.CCR-19-1467](https://doi.org/10.1158/1078-0432.CCR-19-1467)
19. Steven J. Cohen, others. A phase 1b study of erlotinib in combination with gemcitabine and nab-paclitaxel in patients with previously untreated advanced pancreatic cancer. *Cancer*

- Chemotherapy and Pharmacology. 2016;77(4):693–701.
20. Dimitris Karagiannis, Theodoros Rampias. Cancer evolution in precision medicine era. *Cancers (Basel)*. 2022;14(8):1885. doi:[10.3390/cancers14081885](https://doi.org/10.3390/cancers14081885)
 21. Ida Aronchik, others. Mechanisms of resistance to the RAS(ON) multi-selective inhibitor daraxonrasib in pancreatic cancer. ENA Symposium 2025, Revolution Medicines poster; 2025.
 22. Deepak Bhamidipati, others. Impact of FDA project optimus guidance on the design of early-phase clinical trials. *JCO Oncology Advances*. 2025. doi:[10.1200/OA-25-00084](https://doi.org/10.1200/OA-25-00084)
 23. Amber M. Young, others. BATON: Constrained Bayesian optimization for calibrating adaptive trials. 2026.
 24. Daniel D. Von Hoff, others. Increased survival in pancreatic cancer with nab-paclitaxel plus gemcitabine. *New England Journal of Medicine*. 2013;369(18):1691–703. doi:[10.1056/NEJMoa1304369](https://doi.org/10.1056/NEJMoa1304369)
 25. Channing J. Der, Jen Jen Yeh. Advances in RAS therapeutics for pancreatic cancer. *New England Journal of Medicine*. 2026;394(18):1857–61. doi:[10.1056/NEJMe2600517](https://doi.org/10.1056/NEJMe2600517)

Questions